NATIONAL UNIVERSITY OF SCIENCES AND TECHNOLOGY PAKISTAN

School Of Interdisciplinary Engineering & Sciences(SINES)

TBA

Project1

Multi-Omics Integration and Biomarker Discovery using DIABLO

Submitted to: Dr. Masood Ur Rehman

Submitted by: Hafsa Ahmad

Sidra

Mahreen Nawaz

Class: MS-Bioinformatics

a. Background (Brief information of the cancer and dataset)

Breast cancer:

Breast cancer (BRCA) is one of the most prevalent cancers worldwide and a leading cause of cancer-related deaths among women. It is a highly heterogeneous disease at the molecular level, with multiple subtypes that differ in prognosis, treatment response, and survival outcomes. This biological complexity makes integrative, multi-omics analysis crucial for a deeper understanding of its underlying mechanisms and for the identification of biomarkers.

Dataset:

For this project, we used publicly available multi-omics datasets from the **TCGA-BRCA cohort via the Linkedomics portal,** focusing on three layers of data and a metadata file:

- **RNA-Seq (HiSeq, Gene Level):** mRNA expression data for 1,093 tumor samples (20,155 genes), normalized as RPKM.
- **miRNA(HiSeq, Gene Level):**Expression data for 755 samples (823 miRNAs), normalized in RPM.
- **RPPA (Gene Level):** Protein abundance data for 887 samples (175 proteins), measured using Reverse Phase Protein Array.
- **Clinical Data:** Patient-level metadata for 1,097 individuals including 20 clinical attributes such as age, gender, subtype, cancer stage, mortality status, and treatment type etc.

These datasets offer complementary biological insights ranging from gene expression to protein-level regulation and patient phenotypes making them suitable for supervised integration using the DIABLO framework from the **mixOmics** R package. Our goal is to identify multi-omics biomarkers that are predictive of breast cancer subtypes and that contribute to better understanding and classification of the disease.

b. Methods and results(Outlining your steps for analysis)

To identify multi-omics biomarkers associated with breast cancer subtypes, we applied the DIABLO framework (Data Integration Analysis for Biomarker discovery using Latent cOmponents) using the mixOmics package in R. The following are methodological steps:

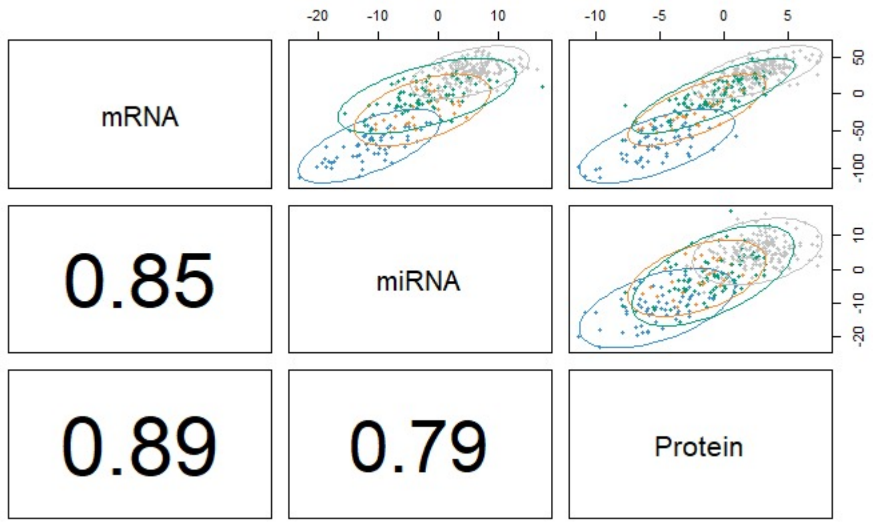
1. **Project Setup and Data Loading**
    i. A R project named **"DIABLO"** was created to manage the working directory and outputs.
    ii. The required omics datasets mRNA, miRNA (843, and RPPA (223 proteins) along with the corresponding clinical data (20 samples) were downloaded from the https://www.linkedomics.org/data_download/TCGA-BRCA/
    iii. The files were loaded into R using **read.table()** with appropriate options for headers and row names.
2. **Preprocessing and Sample Matching**
    i. Since sample identifiers in the omics files used periods (.), we converted them to hyphens (-) using **gsub()** to match the clinical metadata format.
    ii. All omics datasets were transposed so that rows represented samples and columns represented features, aligning with mixOmics requirements.
    iii. We extracted common samples across mRNA, miRNA, RPPA, and clinical data using **intersect()** and filtered each dataset accordingly.
    iv. Only **633 samples** were found to be common across all data layers and were retained for downstream analysis.
3. **Filtering and Normalization**
    i. A **log2 transformation (log2(x + 1))** was applied to each omics dataset to reduce skewness and stabilize variance across features, which is essential for downstream linear modeling.
    ii. Low-variance and constant features were then removed from each dataset using **apply()** with **var()** to retain only informative variables.
    iii. Finally, each omics matrix was scaled and centered using **scale()** to ensure comparability across blocks and to meet the assumptions of the DIABLO model.

4. **Designing and Tuning the Model**
   i. A full design matrix **X** was constructed using **matrix()** to model balanced correlations among the three omics layers.
   ii. The outcome variable **Y** was set as breast cancer **subtype** (Basal, Her2, LumA, LumB) based on the clinical data.
   iii. The optimal number of features per block and component was selected using **tune.block.splsda()** with 10-fold M-fold cross-validation and "centroids.dist" as the distance metric.
5. **Run Diablo and visualization of results**

```
# Step 14: Run DIABLO analysis using block.splsda
diablo_model <- block.splsda(X, Y, design = design)

# Step 15: Visualize the results (optional)
plotDiablo(diablo_model, comp = 1:2)
```

Output:



Interpretation:

This is a correlation matrix and pairwise plot of the first latent components derived from DIABLO across the three omics layers:

Top-right scatter plots show the pairwise distribution of samples for:

mRNA vs miRNA

mRNA vs protein

miRNA vs protein

The ellipses indicate sample grouping and good overlap between ellipses of the same group across plots suggest consistent separation across datasets.

Bottom-left numbers show correlation coefficients between the first components from each data type:

mRNA–miRNA: 0.85

mRNA–protein: 0.89

miRNA–protein: 0.79

These values are high (close to 1), meaning the latent variables extracted from each omics layer are strongly correlated—a good indication that DIABLO has successfully found shared biological signals across the datasets.

What does this mean: The multi-omics data are well-integrated.

There is coordinated regulation between mRNA, miRNA, and proteins. You can use these components to identify biomarkers or classify conditions (like disease vs healthy)

<u>Confusion matrix:</u>

```
# Confusion matrix
conf_matrix <- table(Predicted = pred_classes, Actual = Y_test)
print(conf_matrix)
```

```
> pred_classes <- diablo_pred$class$max.dist
> pred_classes <- as.factor(pred_classes)
Error in order(y) : unimplemented type 'list' in 'orderVector1'
```

Limitations:

Despite successfully running the DIABLO model and obtaining prediction results, we encountered a technical issue while attempting to generate the confusion matrix. Specifically, the output diablo_pred$class$max.dist returned a list instead of a direct vector, causing an error during factor conversion (Error in order(y): unimplemented type 'list' in 'orderVector1'). Although the model was trained and prediction was possible, this structural issue in the output object prevented further evaluation of classification performance. We attempted several debugging steps but were unable to resolve the error within the project's timeline. As a result, visualizations such as the confusion matrix or classification accuracy plots could not be included in this report.

Conclusion

This experience emphasizes the complexity of multi-omics data and the importance of robust preprocessing, variance normalization, and balanced class distribution to ensure meaningful model convergence. With further refinement, this dataset still holds strong potential for biomarker discovery and subtype classification in breast cancer.